

ΨDAG: Projected Stochastic Approximation Iteration for DAG Structure Learning



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Goal

- Learning the data causality structure of the form of Directed Acyclic Graphs (DAGs) is challenging
- Difficulty: highly non-convex nature of DAG constraints

Main idea

- We present a novel framework for learning DAGs, employing a **Stochastic Approximation** approach integrated with Stochastic Gradient Descent (SGD)based optimization techniques
- We enforce DAG constraints efficiently via (new) DAG **projection** methods
- Algorithm **converges to a feasible local minimum**
- Low iteration complexity**, well-suited for handling large-scale problems

Proposed Method

Minimization problem:

$$\min_{w \in \mathbb{R}^d} \left\{ F(w) \stackrel{\text{def}}{=} \mathbb{E}_{\xi} [f(w, \xi)] \right\}$$

The objective function can be expressed as:

$$\min_{\mathbf{W} \in \mathbb{D}} \mathbb{E}_{x \sim \mathcal{D}} \left[l(\mathbf{W}; x) \stackrel{\text{def}}{=} \frac{1}{2} \|x - \mathbf{W}^{\top} x\|^2 = \frac{1}{2} \|x^{\top} - x^{\top} \mathbf{W}\|^2 \right]$$

The proposed framework consists of three main steps:

- Run an optimization algorithm A1 without any DAG constraints, only forcing the diagonal to be zero
- Finding a DAG that is close to the current iterate using projection (A3)
- Run the optimization algorithm A2 while preserving the vertex order

Algorithm 1 ψ DAG framework

- Requires:** Initial model $\mathbf{W}_0 \in \mathbb{R}^{d \times d}$, such that $\text{diag}(\mathbf{W}_0) = 0$.
- for** $k = 0, 1, 2, \dots, K - 1$ **do**
- $\mathbf{W}_k^{(1/3)} = \mathcal{A}_1(\mathbf{W}_k)$
- $(\mathbf{W}_k^{(2/3)}, \pi_k) = \psi(\mathbf{W}_k^{(1/3)})$
- $\mathbf{W}_{k+1} = \mathcal{A}_2(\mathbf{W}_k^{(2/3)}; \pi_k)$
- end for**
- Output:** $\mathbf{W}_K$.

Algorithm 2 ψ DAG

- Requires:** initial model $\mathbf{W}_0 \in \mathbb{R}^{d \times d}$, numbers or iterations τ_1, τ_2 .
- for** $k = 0, 1, 2, \dots, K - 1$ **do**
- $\mathbf{W}_k^{(1/3)} = \text{SGD}(\mathbf{W}_k)$
- $(\mathbf{W}_k^{(2/3)}, \pi_k) = \text{Algorithm 3}(\mathbf{W}_k^{(1/3)})$
- $\mathbf{W}_{k+1} = \text{SGD}_{\pi_k}(\mathbf{W}_k)$
- end for**
- Output:** $\mathbf{W}_K$

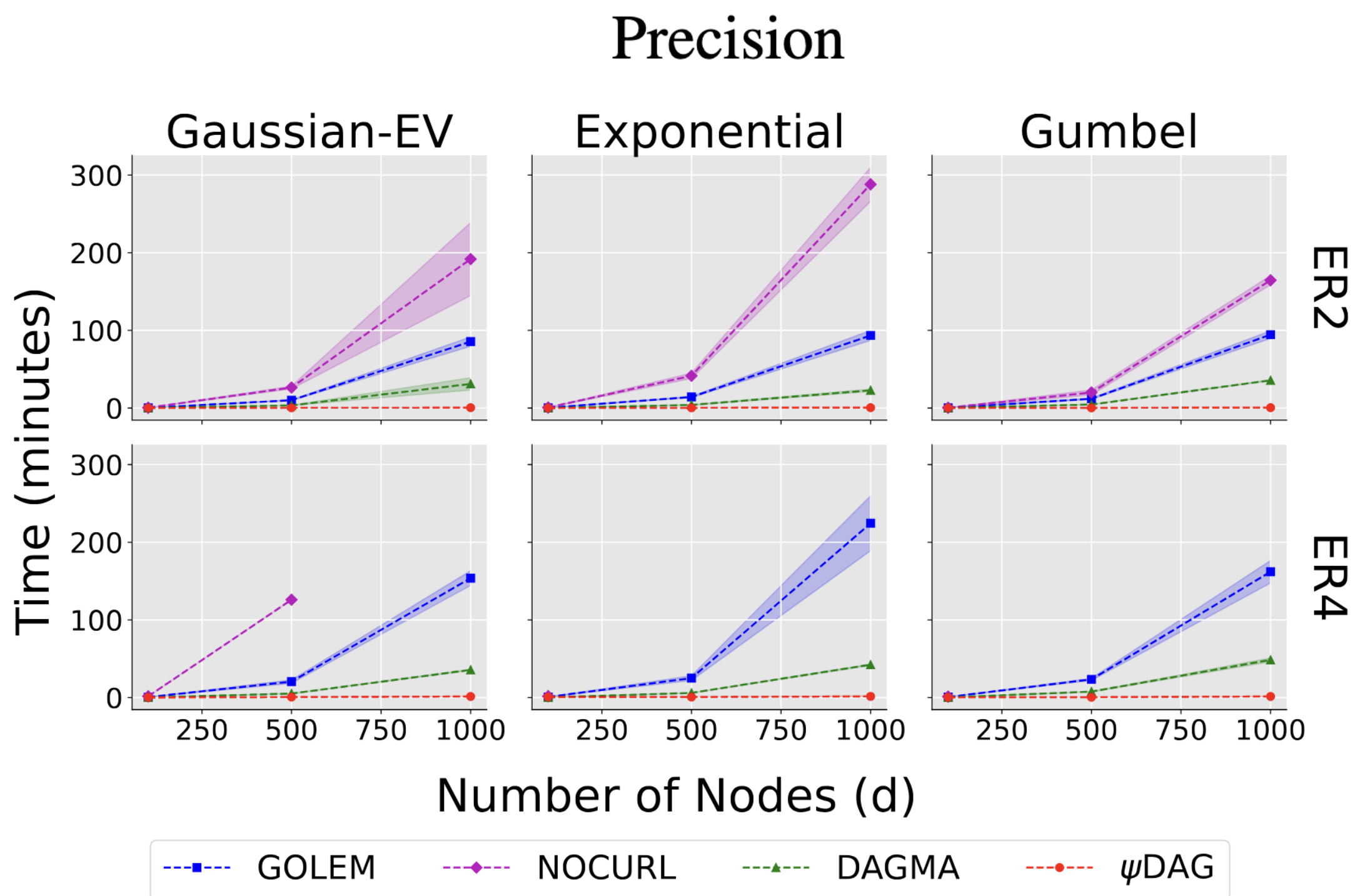
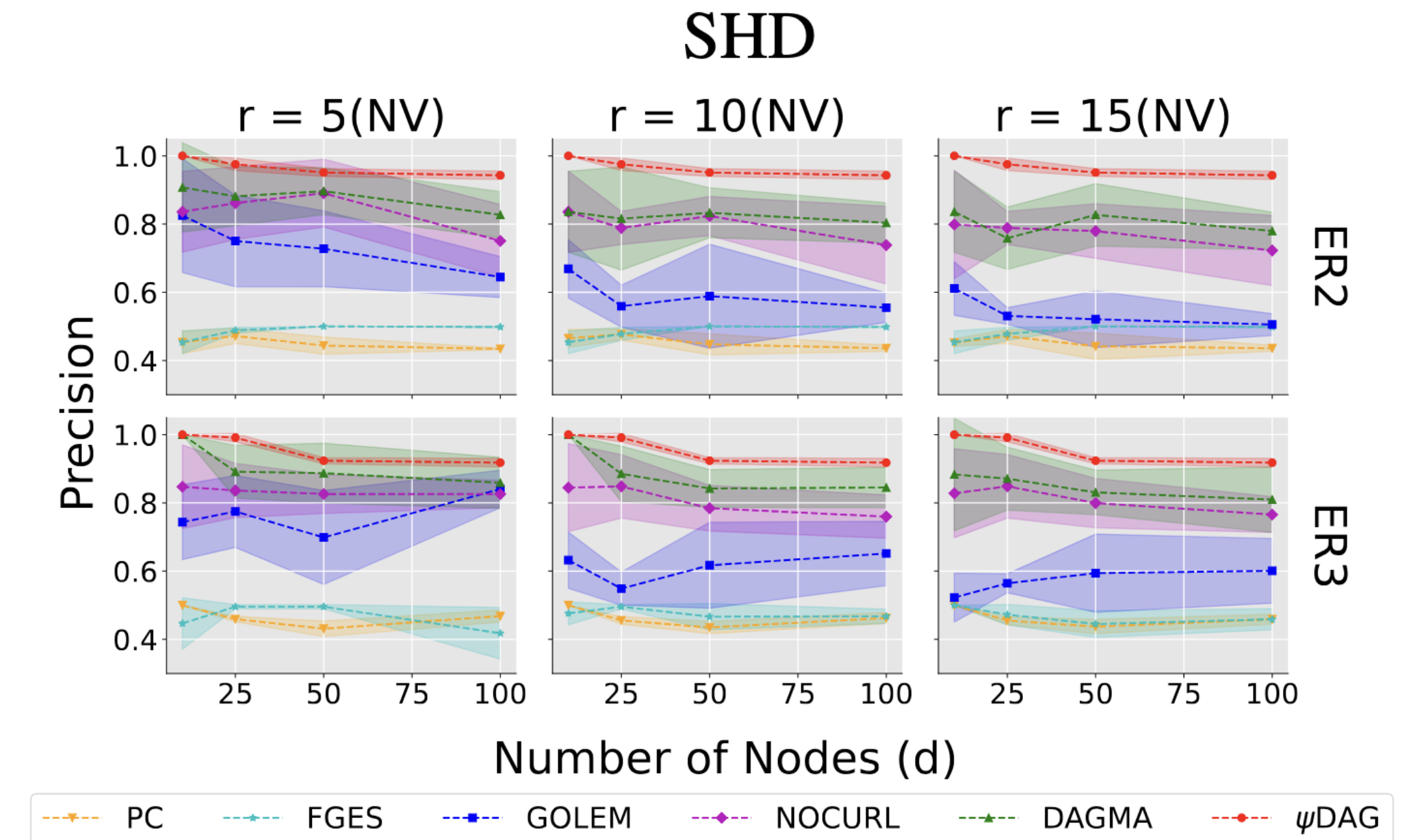
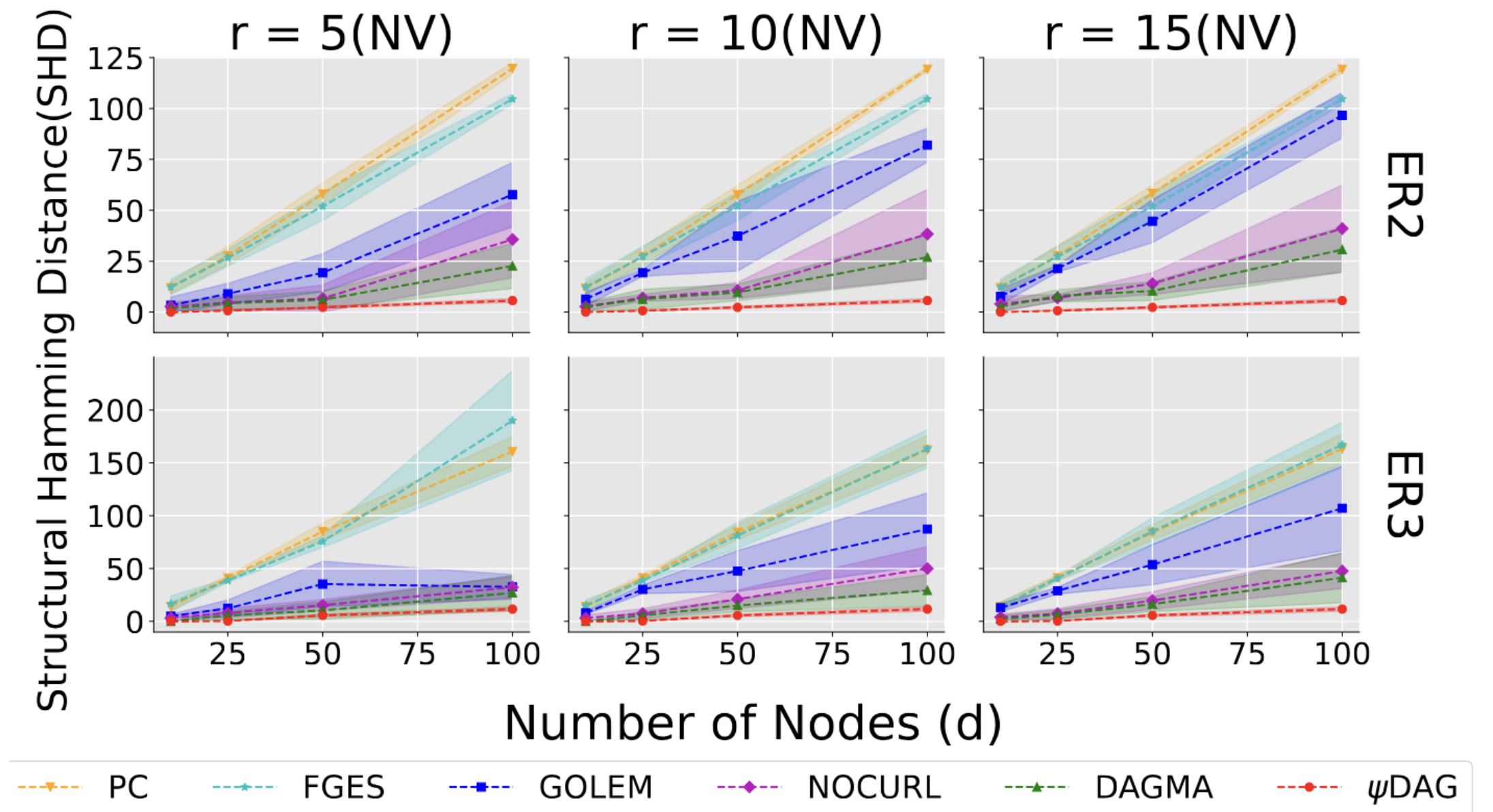
References

- [1] Sachs K., Perez O., Pe'er D., Lauffenburger D., Nolan G.. Causal protein signaling networks derived from multiparameter single-cell data. Science 2005
- [2] Ng I., Ghassami A., and Zhang K. On the role of sparsity and DAG constraints on learning linear DAGs. NeurIPS 2020
- [3] Zheng X., Aragam B., Ravikumar P., Xing E.. DAGs with no tears: Continuous optimization for structure learning. NeurIPS 2018
- [4] Bello K., Aragam B., Ravikumar P. DAGMA: Learning DAGs via M-matrices and a log-determinant acyclicity characterization. NeurIPS 2022

Algorithm 3 Projection $\psi(\mathbf{W})$ computing the “closest” vertex ordering (recursive form)

- Requires:** Model $\mathbf{W} \in \mathbb{R}^{d \times d}$, (optional) weights $\mathbf{L} \in \mathbb{R}^{d \times d}$ with default value $\mathbf{L} = \mathbf{11}^{\top}$.
 - for** $k = 1, \dots, d$ **do**
 - Set $r_k = \|(\mathbf{W} \circ \mathbf{L})[k][:]\|^2$
 - Set $c_k = \|(\mathbf{W} \circ \mathbf{L})[:,k]\|^2$
 - end for**
 - Set $i_c = \arg \min_{k \in \{1, \dots, d\}} c_k$
 - Set $i_r = \arg \min_{k \in \{1, \dots, d\}} r_k$
 - if** $r_{i_r} \leq c_{i_c}$ **then**
 - Output:** $[\psi(\mathbf{W}(i_c, i_c), \mathbf{L}(i_c, i_c)), i_r]$
 - else**
 - Output:** $[i_c, \psi(\mathbf{W}(i_c, i_c), \mathbf{L}(i_c, i_c))]$
 - end if**
- {By $A(i, j)$ we denote the submatrix $A[1, \dots, i - 1, i + 1, \dots, d][1, \dots, j - 1, j + 1, \dots, d]$ }

Results



Large Graph

